These are notes for Malek Abdesselam's Functional Analysis class for Fall 2026.

Contents

Motivation

Consider a space V of (sufficiently nice) functions $f: \mathbb{R} \to \mathbb{R}$. Then, similar to how a vector in $\mathbb{R}^n$ can be denoted as

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may view the function $f: \mathbb{R} \to \mathbb{R}$ as a "column vector" with uncountable index. That is, $f = (f(x))_{x \in \mathbb{R}}$. If $L\colon V \to V$ is a linear map, then an analogue of the finite-dimensional matrix representation can be viewed as a kernel that f is integrated against; that is,

$$(Lf)(x) = \int_{\mathbb{R}} K(x,y) f(y)\, dy.$$

In the special case that $L = \mathrm{id}_V$, then the kernel $K(x,y)$ is given by the Dirac delta distribution, $K(x,y) = \delta(x-y)$. The space of these distributions is the primary object of study in Fourier analysis.

Locally Convex Topological Vector Spaces

Recall that if X is a set, then $\mathcal{T} \subseteq \mathcal{P}(\mathcal{P}(X))$ is a topology on X if it satisfies:

- $\emptyset, X \in \mathcal{T}$;
- $U_1, U_2 \in \mathcal{T}$ implies $U_1 \cap U_2 \in \mathcal{T}$;
- $\{U_i : i \in I\} \subseteq \mathcal{T}$ implies $\bigcup_{i\in I} U_i \in \mathcal{T}$.

Definition: **Definitions**

We say $(V, \mathcal{T})$ is a *topological vector space* (over $\mathbb{K}$) if V is a $\mathbb{K}$-vector space whose topology is such that the maps

$$\begin{aligned} V \times V &\to V \\ (u,v) &\mapsto u+v \end{aligned}$$

and

$$\begin{aligned} \mathbb{K} \times V &\to V \\ (\lambda,v) &\mapsto \lambda v \end{aligned}$$

are continuous with respect to the product topology. Note that $\mathbb{K}$ is equipped with its traditional topology.

The morphisms of topological vector spaces are the *continuous* linear maps. We say $V \cong W$ are isomorphic as topological vector spaces if there is $\varphi\colon V \to W$ that is bijective, linear, and continuous, and φ^{-1} is also continuous.

Definition: Let V be a $\mathbb{K}$-vector space. We say $\rho\colon V \to [0,\infty)$ is a *seminorm* if

$$\rho(v+w) \leq \rho(v) + \rho(w)$$
$$\rho(\lambda v) = |\lambda|\rho(v)$$

for any $v, w \in V$ and $\lambda \in \mathbb{K}$.

Remark: The only difference between a seminorm and a norm is that $\rho(v) = 0$ need not imply that $v = 0$. A norm is a seminorm.

We will let $\mathrm{SN}(V)$ be the set of all seminorms of V.

Definition: Let V be a $\mathbb{K}$-vector space. Let $\mathcal{N} \subseteq \mathrm{SN}(V)$. Define $\mathcal{T}(\mathcal{N})$ to be a topology with basic open sets $U(x,F,\varepsilon) \subseteq V$ given by

$$U(x,F,\varepsilon) = \{y \in V : \rho(y-x) < \varepsilon \text{ for all } \rho \in F\}$$

for any $x \in V$, $F \subseteq \mathcal{N}$ finite, and $\varepsilon > 0$.

We observe that the subsets of the form $U(x,\{\rho\},\varepsilon)$ as ρ ranges across $\mathcal{N}$ forms a subbase for the topology of V.

Proposition: If $F \subseteq \mathcal{N}$ is finite, $\varepsilon > 0$, and $x \in V$, then

$$U(x,F,\varepsilon) \in \mathcal{T}(\mathcal{N}).$$

Proof. Suppose $y \in U(x,F,\varepsilon)$. Our goal is to find $\eta > 0$ such that $U(y,F,\eta) \subseteq U(x,F,\varepsilon)$. Suppose $z \in U(y,F,\eta)$. If $\rho \in F$, then

$$\begin{aligned}\rho(z-x) &\leq \rho(z-y) + \rho(y-x)\\ &< \eta + \rho(y-x).\end{aligned}$$

Therefore, we will define η to be small enough such that $\eta + \max_{\rho\in F}\rho(y-x) < \varepsilon$. This gives the desired result. □

Definition: A topological space is called a locally convex topological vector space (LCTVS) if the topology is generated by a family of seminorms $\mathcal{N}$ with the property that

$$\bigcap_{\rho\in\mathcal{N}}\{v : \rho(v) = 0\} = \{0\}.$$

A family $\mathcal{N}$ of seminorms that satisfies this property is said to be *separating*.

Example: If H is a Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$, then the seminorms $v \mapsto |\langle v, e_i\rangle|$ are separating.

Theorem: If V is a $\mathbb{K}$-LCTVS and $\mathcal{N}$ is a collection of seminorms that defines the topology on V, then $(V,\mathcal{T}(\mathcal{N}))$ is Hausdorff if and only if $\mathcal{N}$ separates the points of V.

Proof. Assume $\mathcal{N}$ separates the points of V. Suppose $v \neq w$ are distinct elements of V. Then, since $v - w \neq 0$, there is some $\rho \in \mathcal{N}$ such that $\rho(v-w) = \varepsilon_0 > 0$. Consider the finite set $F = \{\rho\}$, and choose some $0 < \varepsilon < \varepsilon_0/2$. Then, we claim that □

Examples

Remark: If we remove the assumption that the family $\mathcal{N}$ is separating, then the topology induced by the trivial seminorm $\rho\colon V \to [0,\infty)$ is the indiscrete topology on V. Similarly, there is a family of seminorms that induces the discrete topology on V.

(i) Let $(V,\|\cdot\|)$ be a normed vector space over $\mathbb{K}$. Examples of these include Banach spaces such as

$L^p(X,\Sigma,\mu)$, where Σ is a σ-algebra and μ is a positive measure on the measurable space (X,Σ), and $p \in [1,\infty]$.

There is a metric on this normed space $d(x,y) = \left\|x-y\right\|$, which gives rise to the metric topology $\mathcal{T}(d)$. We have $U \in \mathcal{T}(d)$ if and only if $U \subseteq V$ is such that for any $x \in U$, there exists $\varepsilon > 0$ such that

$$\begin{aligned}\{y \in V : d(x,y) < \varepsilon\} &=: U(x,\varepsilon)\\ &\subseteq U.\end{aligned}$$

This is a locally convex space with seminorm $\mathcal{N} = \{\|\cdot\|\}$. There are two possibilities for finite subsets $F \subseteq \{\|\cdot\|\}$, meaning that if $F = \emptyset$, then $U(x,F,\varepsilon) = U(x,\emptyset,\varepsilon) = V$, while if $F = \{\|\cdot\|\}$, then

$$\begin{aligned}U(x,F,\varepsilon) &= \{y \in V : \rho(x-y) < \varepsilon \text{ for all } \rho \in F\}\\ &= \left\{y \in V : \left\|x-y\right\| < \varepsilon\right\}\\ &= U(x,\varepsilon).\end{aligned}$$

(ii) Define $s(\mathbb{N},\mathbb{K})$ to be the space of sequences $x = (x_n)_{n\geq 0}$ in $\mathbb{K}^{\mathbb{N}}$ such that for all $k \geq 0$, $\sup_{n\geq 0}\langle n\rangle^k |x_n| < \infty$, where $\langle n\rangle = \sqrt{1+n^2}$ is the bracket. This is the space of *rapidly decaying sequences* in $\mathbb{K}$. In essence, elements of $s(\mathbb{N},\mathbb{K})$ decay faster than any inverse power of the index n.

We define the seminorms

$$\rho_k(x) = \sup_{n\geq 0}\langle n\rangle^k |x_n|.$$

In fact, the $\{\rho_k : k \in \mathbb{N}\}$ are norms. Let

$$\mathcal{N} := \{\rho_k : k \in \mathbb{N}\}$$

We obtain a locally convex on $s(\mathbb{N},\mathbb{K})$ with topology induced by the family $\mathcal{N}$.

Later, we will see that $(s(\mathbb{N},\mathbb{K}),\mathcal{T}(\mathcal{N}))$ is *not* a normed space. That is, there does not exist a norm $\|\cdot\|$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\{\|\cdot\|\})$.

(iii) Take $V = C([0,\infty),\mathbb{R})$ to be the set of all real-valued continuous functions on $[0,\infty)$ with pointwise addition and scalar multiplication. This is a real vector space.

At first, we may think $\|f\| = \sup_{t\in\mathbb{R}}|f(t)|$ would define a norm, but this is not satisfactory since the functions in V are not necessarily bounded.

Therefore, we want to restrict to compact subsets. Take

$$\rho_k(f) := \sup_{0\leq t\leq k}|f(t)|.$$

Notice that the ρ_k are not norms (use Urysohn's Lemma or your favorite piecewise function to construct one). However, the family $\mathcal{N} = \{\rho_k : k \geq 0\}$ is a separating family.

Consider the set of functions $V_0 = \{f \in V : f(0) = 0\}$. Then, $V_0 \leq V$ is a closed subset.

Proof. □

We claim that the subspace topology on V_0 is given by $\mathcal{T}_0 = \mathcal{T}\left(\left\{\rho_k \bigm|_{V_0} : k \geq 0\right\}\right)$.

Proof. □

We may consider $\mathcal{B}(\mathcal{T}_0)$ to be the Borel σ-algebra on V_0. Brownian motions are a special class of probability measure on $(V_0,\mathcal{B}(\mathcal{T}_0))$. Specifically, they are the probability measures such that for every $n \geq 0$, partition $0 < t_1 < t_2 < \cdots < t_n$, the random variables

$$X_1 = f \mapsto f(t_1)$$

$$\begin{aligned} X_2 &= f \mapsto (f(t_2) - f(t_1)) \\ X_3 &= f \mapsto (f(t_3) - f(t_2)) \\ &\vdots \\ X_n &= f \mapsto (f(t_n) - f(t_{n-1})) \end{aligned}$$

are independent centered Gaussian random variables with variances $t_1, t_2 - t_2, \dots, t_n - t_{n-1}$ for an arbitrary f.

Similar to the case of $s(\mathbb{N}, \mathbb{K})$, the space $(V_0, \mathcal{T}_0)$ is not a normed space.

(iv) Consider the vector space $V = \mathbb{R}^{\mathbb{N}}$, which is the space of all real-valued sequences. Let $\mathcal{T}$ be the product topology on $\mathbb{R}^{\mathbb{N}}$. We are curious as to whether $(V, \mathcal{T})$ is a locally convex topological vector space.

Consider the function $d \colon V \times V \to [0, \infty)$ defined by

$$d(x, y) = \sum_{n=0}^{\infty} 2^{-n} \min\left(1, \left|x_n - y_n\right|\right).$$

Then, we claim that d is a distance metric, and $\mathcal{T} = \mathcal{T}(d)$.

Proof. □

In particular, we may consider the seminorms

$$\rho_n(x) = |x_n|,$$

and $\mathcal{N} = \{\rho_n : n \geq 0\}$. Then, we claim that $\mathcal{T} = \mathcal{T}(\mathcal{N})$.

Proof. □

In particular, V is not a normed space.

(v) Consider $c_{00}(\mathbb{N}, \mathbb{K}) \cong \mathbb{K}[x]$ to consist of all the finitely supported sequences $(x_n)_{n \geq 0}$. This is a vector space. Consider the seminorms $\mathcal{N} = \mathrm{SN}(V)$, and the locally convex space $(V, \mathcal{T}(\mathcal{N}))$. Then, this is a locally convex space that is not metrizable. The topology $\mathcal{T}(\mathcal{N})$ is known as the finest locally convex topology.

Notation

- The natural numbers are given by $\mathbb{N} = \{0, 1, 2, 3, \dots\}$.
- The strictly positive natural numbers are given by $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$.
- A multi-index is a tuple $\alpha \in \mathbb{N}^n$. We let

  $$|\alpha| = \alpha_1 + \cdots + \alpha_n.$$

  If $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, then we will write

  $$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

  and

  $$\alpha! = \alpha_1! \cdots \alpha_n!$$

  for the corresponding values.

  If $f \colon \mathbb{R}^n \to \mathbb{R}$ is any function, then we will write

  $$\partial^\alpha f = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

- If A is a set, we will write $\mathcal{P}(A)$ for the power set of A. If A and B are sets, then the set $B^A \subseteq \mathcal{P}(A \times B)$ is the set of all functions $f : A \to B$.

  We will write $\mathcal{P}_{\text{fin}}(X)$ to consist of all finite subsets subsets of X.

- If $x \in \mathbb{R}^n$ is given by $(x_1, \dots, x_n)$, we will define the bracket

$$\langle x \rangle := \sqrt{1 + x_1^2 + \cdots + x_n^2}.$$

  In the particular case of $n = 1$, we will write $\langle k \rangle = \sqrt{1 + k^2}$ for any $k \in \mathbb{N}$.

- If $n \in \mathbb{N}$, then

$$[n] = \{1, 2, \dots, n\},$$

  where we have $[0] = \emptyset$.

- If A is a finite set, then $|A|$ is the cardinality of A.

- Generally we will denote by $\mathbb{K}$ for either $\mathbb{R}$ or $\mathbb{C}$.